

CN4119 Final Year Design Project

Cost Estimation – Capital Cost

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Recommended Readings

Turton et al.'s book (Reference 5) Chapter 7

Overview

- ❑ Types of capital cost estimates
- ❑ Purchased equipment cost
- ❑ Cost index (to account for time)

- ❑ Capital cost estimation of a plant - ***2 methods***:
 - Six-tenths rule
 - Module costing technique

- ❑ Summary and ***software***

Types of Estimates

Types of Estimates

- ❑ Approximate to accurate estimates
- ❑ Minimal to extensive data/effort
- ❑ 5 types of cost estimates *(next slide)*
- ❑ *Most accurate* estimate: detailed or firm or contractor's
- ❑ *Least effort* estimate: order-of-magnitude or ratio

Type	Purpose	Accuracy	Cost
Order-of-magnitude or ratio	Screening or feasibility	4 to 20*	<i>0.015% to 0.3% of plant cost</i>
Study or factored or major equipment	Concept study or feasibility	3 to 12*	2 to 4 [#]
Preliminary design or scope	Budget, approval or control	2 to 6*	3 to 10 [#]
Definitive or project control	Control or bid/tender	1 to 3*	5 to 20 [#]
Detailed or firm or contractor's	Check estimate or bid/tender	<i>Error: +6% to - 4%</i>	10 to 100 [#]

* Relative to accuracy of the detailed estimate

Relative to cost of the order-of-magnitude estimate

Note: Base estimate equivalent to 1

Increasing
accuracy

Decreasing
cost

Purchased Equipment Cost

Purchased Equipment Cost

- ❑ Capital cost estimation requires ***purchase cost of major equipment*** in the plant

- ❑ ***Effect of size (or capacity)***

Purchase cost of equipment, C correlated with its size, S via the exponential equation:

$$\frac{C}{C_r} = \left(\frac{S}{S_r} \right)^n$$

- Subscript $r \rightarrow$ reference/base case
- Exponent $n \rightarrow 0.3$ to 0.84
- Value of $n \approx 0.60$ for most equipment $\Rightarrow$ ***six-tenths rule***

- ❑ Re-writing previous equation:

$$C = K(S)^n$$

- K in terms of C_r and S_r ?

- Plot $\log C$ versus $\log S \Rightarrow$ straight line with slope n

□ Typical values of cost exponents

Equipment	Size Range with Units	n	C_r in US\$ (S_r in brackets)
Reciprocating compressor (motor drive)	0.75 to 1490 kW	0.84	133,000 (224 kW)
Heat exchanger (shell and tube, carbon steel)	1.9 to 1860 m ²	0.59	21,700 (9.3 m ²)
Centrifugal blower (excluding motor)	0.24 to 71 std m ³ /s	0.60	67,000 (4.72 m ³ /s)
Jacketed kettle (glass lined)	0.2 to 3.8 m ³	0.48	53,000 (0.38 m ³)

- Now, cost correlation

$$C = K(S)^n$$

can be rearranged to:

$$\frac{C}{S} = K(S)^{n-1}$$

- Effect of increasing size (S) on the ***equipment cost per unit capacity?***

- Economy of scale

- ❑ **Cost plots:** Example – Heat exchanger purchase cost from Figure A.5 in Turton's book (2013 edition)

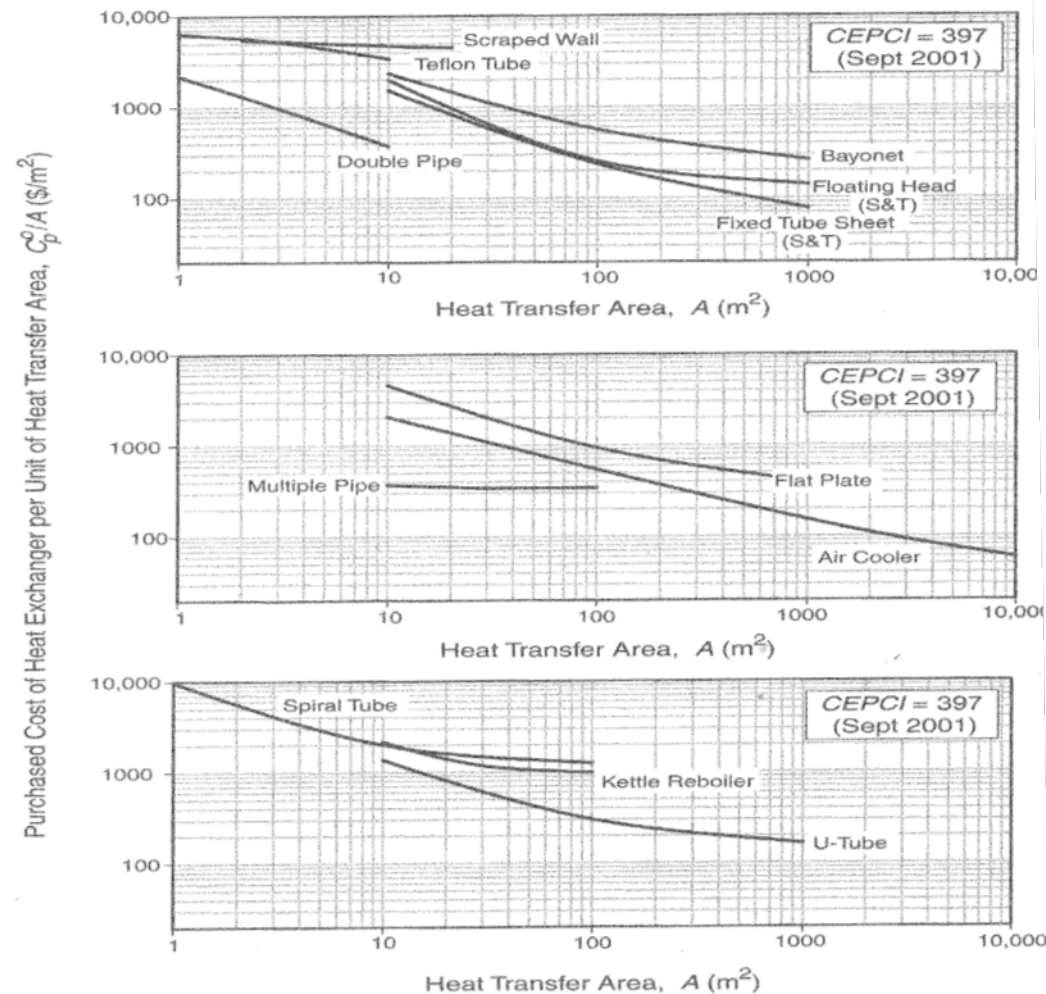


Figure A.5 Purchased Costs for Heat Exchangers

Test Your Understanding

Example 7.3: Economy of Scale

Use six-tenths rule to estimate the percentage increase in purchase cost when the capacity of a piece of equipment is doubled.

- ❑ Application of exponential equation: Cost of an equipment increases by 52% when its size doubles

Test Your Understanding

Example 7.4: Effect of Exponent n

Compare the error for the scale-up of a reciprocating compressor by a factor of 5 using the six-tenths rule in place of the cost exponent for a reciprocating compressor.

- ❑ Percentage error is -32%

Example 7.5: Application of the Exponential Equation

The purchase cost of a recently acquired heat exchanger of area 100 m^2 is \$10,000.

Determine constant K and hence the cost of a new heat exchanger of area 180 m^2 .

- The current purchase cost presents the '*base case*' information
- K is $661 \text{ } \$/(\text{m}^2)^{0.59}$ and new heat exchanger cost is \$14,100

$$K = \frac{C_r}{S_r^n} =$$

$$C = K(S)^n =$$

Cost Index

Cost Index

- ❑ Equipment cost varies with time due to inflation and technological developments
- ❑ Historical cost may have to be used to estimate equipment cost in the next few months to years

=> Suitable cost index (similar to *Consumer Price Index, CPI*) required to update historical equipment cost
- ❑ Cost, C in certain year with cost index = I given by:

$$\frac{C}{C_b} = \frac{I}{I_b}$$

➤ Here, C_b is the known cost in the year when the index was I_b

❑ Cost indexes used by the chemical industry

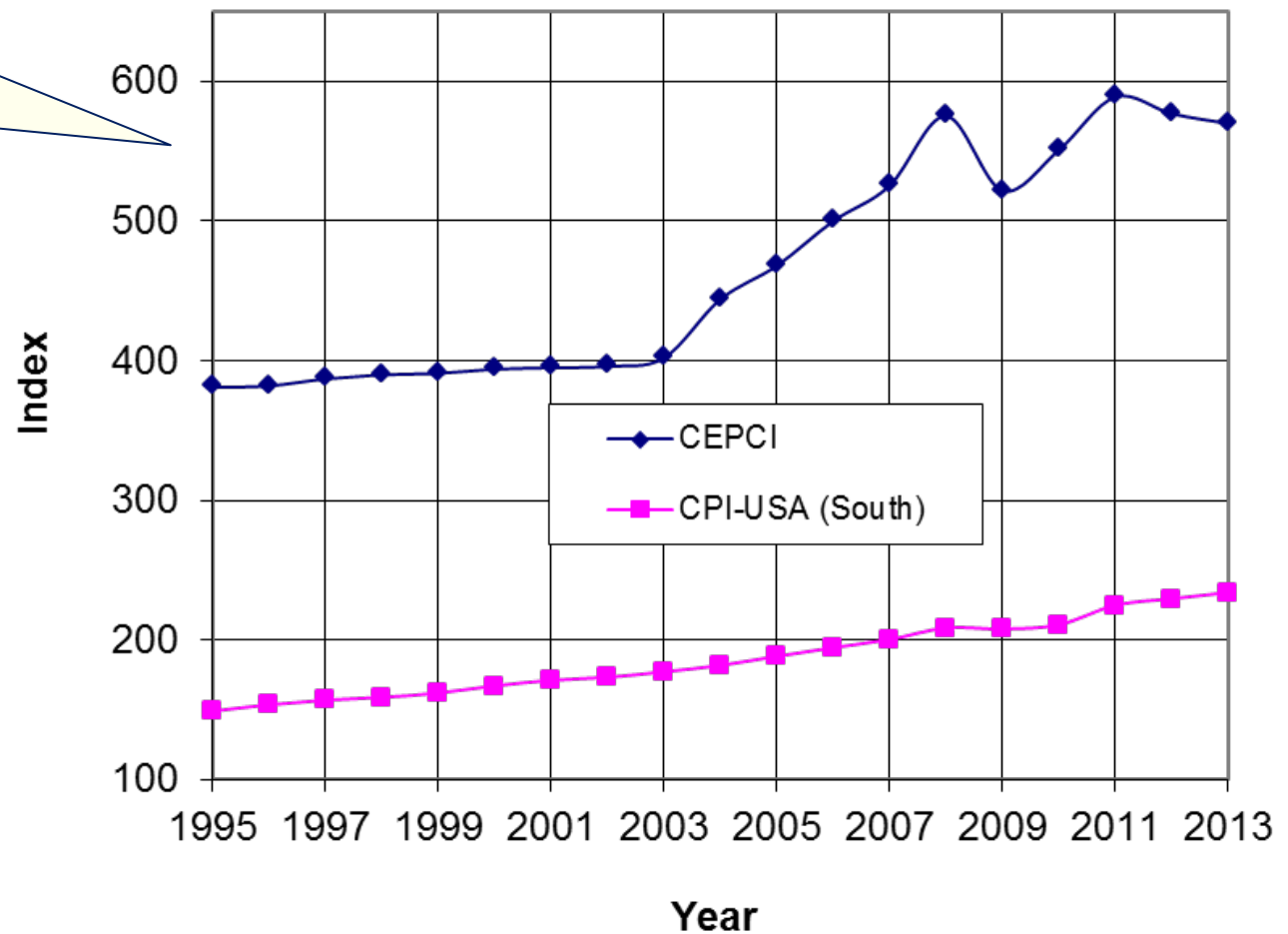
Cost Index, I	Year when I = 100	Value in Year 2000
Chemical Engineering Plant Cost Index, CEPCI	1957	394
Engineering News-Record Construction Index	1967	579
Marshall & Swift Equipment Cost Index	1926	1089
Nelson-Farrar Refinery Construction Index	1946	1543

CEPCI available in the *Chemical Engineering* technical magazine (<http://www.chemengonline.com/the-magazine/>) or research articles

CEPCI α *Composite Index* consisting of:

No.	Component/Item	Weight
1	Fabricated equipment	0.2257
2	Process machinery	0.0854
3	Pipes, valves and fittings	0.122
4	Process instruments and controls	0.0427
5	Pumps and compressors	0.0427
6	Electrical equipment and materials	0.0305
7	Structural supports, insulation and paint	0.061
8	Erection and installation labor	0.22
9	Buildings, materials and labor	0.07
10	Engineering and supervision	0.10

Variation of CEPCI and CPI from 1995 to 2013



- ❑ Do you need CEPCI for the future?
 - If required, extrapolate with caution

What Constitutes Capital Cost of a Plant?

- ❑ Capital cost of a plant >>> Sum of purchase cost of all equipment

Factor (<i>See book for details</i>)	Cost Symbol/Equation
(a) Equipment purchase (b) Materials for installation (c) Labor for installation <i>Direct Project Expense</i>	C_P $C_M = \alpha_M C_P$ $C_L = \alpha_L (C_P + C_M)$ $C_{DE} = C_P + C_M + C_L$
(a) Freight, insurance & taxes (b) Construction overhead (c) Contractor engineering expenses <i>Indirect Project Expense</i>	$C_{FIT} = \alpha_{FIT} (C_P + C_M)$ $C_O = \alpha_O C_L$ $C_E = \alpha_E (C_P + C_M)$ $C_{IDE} = C_{FIT} + C_O + C_E$
Bare Module Cost	$C_{BM} = C_{DE} + C_{IDE}$

This and the next slide conveys **“What is capital cost”**.
“How to calculate capital cost?” is covered later from Slide 24.

Factor (<i>See book for details</i>)	Cost Symbol/Equation
Bare Module Cost	$C_{BM} = C_{DE} + C_{IDE}$
(a) Contingency (b) Contractor fee	$C_{Cont} = \alpha_{Cont} C_{BM}$ $C_{Fee} = \alpha_{Fee} C_{BM}$
Total Module Cost	$C_{TM} = C_{BM} + C_{Cont} + C_{Fee}$
Total module cost for developed sites and additions to existing plants	
<i>Auxiliary facilities</i> (a) Site/land and its development (b) Auxiliary building (c) Off-sites and utilities	$C_{Site} = \alpha_{Site} C_{BM}$ $C_{Aux} = \alpha_{Aux} C_{BM}$ $C_{Off-sites} = \alpha_{Off-sites} C_{BM}$
Grass Roots Cost	$C_{GR} = C_{TM} + C_{Site} + C_{Aux} + C_{Off-sites}$
Grass roots cost for green field sites and new plants	

- ❑ Values of α 's depend on equipment
- ❑ We can compute a multiplication factor to compute the bare module cost:

$$F_{BM} = (1 + \alpha_M)(1 + \alpha_L + \alpha_{FIT} + \alpha_L \alpha_O + \alpha_E)$$

- ❑ F_{BM} values available in Appendix A of Turton's book

What We Have Covered So Far!

- ❑ Cost estimates vary in terms of **effort (time/money)** and **accuracy**
- ❑ **Six-tenths rule** to account for effect of capacity on cost
- ❑ **Cost index** to account for effect of time
- ❑ **Purchase cost** correlations given as C/S vs S on log-log scale
- ❑ **Capital cost** of a plant

Direct project expense + indirect project expense

$$= BMC \xrightarrow{+(contingency \& contractor \ fee)} TMC \xrightarrow{+(auxiliary \ facilities)} GRC$$

Capital Cost of a Plant

- 1. Six-Tenths Rule**
- 2. Module Costing Technique**

1. Six-Tenths Rule

- ❑ Exponential equation with $n = 0.6$ (six-tenths) for estimating capital cost of a **plant** (or an equipment) ***approximately and quickly***:

$$\frac{C}{C_r} = \left(\frac{S}{S_r} \right)^{n=0.6}$$

- Six-tenths rule more accurate here than for estimating cost of a single piece of equipment
- **Why?**

Example 7.7: Six-Tenths Rule

Capital cost of a plant for producing 30,000 tons/year of isopropanol in 1996 (CEPCI = 382) was \$23 million. Estimate the capital cost for setting up a new plant to produce 50,000 tons/year of isopropanol in 2011 (CEPCI = 582).

- ❑ Estimated capital cost in 2011 is equal to \$47,610,000
- ❑ 2 steps: Use six-tenths rule to estimate capital cost of 50,000 tons/year plant in 1996 itself, then use CEPCI to update this cost to the year 2011

2. Module Costing Technique

- Introduced by Guthrie around 1970
- Basis for many techniques used today
 - Relates all costs back to the purchase cost of equipment evaluated for some ***base conditions***
 - Deviations from these base conditions handled by using ***multiplying factors***:

$$C_{BM} = C_p^0 F_{BM} = C_p^0 (B_1 + B_2 F_p F_M) \text{-----}(1)$$

Where,

C_p^0 = Purchase cost of equipment at base conditions (***carbon steel material and ambient pressure operation***)

B_1 and B_2 = Coefficients, whose values depend on the equipment

F_p and F_M = Pressure and material factors

F_{BM} = Bare module factor

2a. Purchase Cost

- Correlations/plots for C_p^0 of different equipment in Appendix A of Turton et al.'s book:

$$\log_{10} C_p^0 = K_1 + K_2 \log_{10}(S) + K_3 [\log_{10}(S)]^2 \text{ -----(2)}$$

Where

S = Capacity/size of the equipment

K_i = Coefficients given in Table A.1 of Appendix A

- *Use the correlation/plot within the range stated*
- *Avoid extrapolation*

2b. Pressure Factor

□ Pressure Factor, F_p

- *Accounts for deviation in pressure from base value of 1 atm*
- $\uparrow$ pressure (or vacuum) $\Rightarrow$ $\uparrow$ equipment cost due to $\uparrow$ wall thickness
- *Always use maximum pressure for pressure factor determination*
- Pressure factor for equipment **other than process vessels (i.e., pumps, heaters, exchangers, etc.)**:
 - In general, pressure factor correlated with P as follows:

$$\log_{10} F_p = C_1 + C_2 \log_{10}(P) + C_3 [\log_{10}(P)]^2 \text{-----}(3)$$

Values of coefficients, C_n given in Table A.2 of Appendix A

■ Pressure factor for ***process vessels (horizontal/vertical)***:

- For equipment operating below -0.5 *barg*, vessel designed to withstand full vacuum, that is, 1 *bar* of external pressure, $F_p = 1.25$
- For equipment operating above -0.5 *barg* and $t < t_{min}$, $F_p = 1$
- For equipment operating above -0.5 *barg* and $t > t_{min}$,

$$F_p = \frac{\frac{PD}{2(850) - 1.2P} + CA}{t_{min}} \text{-----(4)}$$

Where:

D = vessel diameter in *m*

P = operating pressure in *barg* (***must use maximum pressure in the unit***)

CA = corrosion allowance (assumed to be 0.00315 *m*, that is, 1/8 *in.*)

t_{min} = minimum allowable vessel thickness (assumed to be 0.0063 *m*, that is, ¼ *in.*)

S = maximum allowable stress = 944 *bar* for carbon steel

E = welding efficiency = 0.9

- F_p is the ***ratio of wall thickness at operating condition to minimum allowable wall thickness***
 - ✓ Wall thickness increases with increasing operating pressure
 - ✓ If calculated $F_p < 1$, we make $F_p = 1$ to be conservative
 - ✓ Minimum wall thickness to give sufficient rigidity to vessels
- Corrosion allowance is included even if there is no concern of corrosion
- S is the maximum allowable stress (force per unit area) that a material can withstand
 - ✓ It has units of pressure (bar)
 - ✓ Wall thickness $\uparrow$ as $S \downarrow$
- Vessels are fabricated by welding
 - ✓ E accounts for the efficiency of the welding
 - ✓ Wall thickness $\uparrow$ as $E \downarrow$

2c. Material Factor

□ Material Factor, F_M

- *Accounts for deviation of material from base material of carbon steel*
- Choice of material of construction (MOC) depends on the chemicals that will contact the equipment walls
- Different MOC needed to meet budget, corrosion, operating temperature and operating pressure requirements
- F_M accounts for the cost of different MOC
- F_M for different materials and equipment given in Figure A.18 and Table A.3

MOC used in process industry

Material	Cost	Comments
Carbon Steel	Base Case	<i>Less than 1.5 wt % carbon, most economical and common, MOC when corrosion is not a concern</i>
Low-alloy Steel	Low-Moderate	With chromium (for resistance to mildly acidic & oxidizing conditions) and molybdenum (for strength at high T)
Stainless Steel	Moderate	<i>More than 12 wt % chromium, high resistance to chemicals and rusting</i>
Aluminum & its alloys	Moderate	High strength-to-weight ratio, good corrosion resistance
Copper & its alloys	Moderate	High thermal conductivity, resistance to seawater
Titanium & its alloys	High	Good strength-to-weight ratio and resistant to oxidizing agents
Nickel & its alloys	High	Nickel with copper (<i>Monel</i>), chromium (<i>Inconel</i>) and molybdenum (<i>Hastelloy</i>), excellent chemical resistance at high T

Example 7.12: BMC

Calculate the bare module cost of a 100 m^2 floating-head shell-and-tube heat exchanger for the following cases (MOC stainless steel):

- 1) Operating pressure on both shell and tube sides = 1 barg
- 2) Operating pressure on both shell and tube sides = 100 barg

Assume $C_P^0 = \$36,650$

- For 1), $F_P = 1$ and $F_M = 2.73 \Rightarrow C_{BM}$ (using Equation 1) = \$225,800
- For 2), $F_P = 1.383$ and $F_M = 2.73 \Rightarrow C_{BM} = \$289,300$
- The effect of pressure on bare module cost of a heat exchanger

$$C_{BM} = C_P^0 F_{BM} = C_P^0 (B_1 + B_2 F_P F_M)$$

Equipment coefficients:

Pressure factor:

$$\log_{10} F_P = C_1 + C_2 \log_{10}(P) + C_3 [\log_{10}(P)]^2$$

P = 1 barg:

P = 100 barg:

Material factor:

$$P = 1 \text{ barg: } C_{BM} = C_P^0 F_{BM} = C_P^0 (B_1 + B_2 F_P F_M)$$

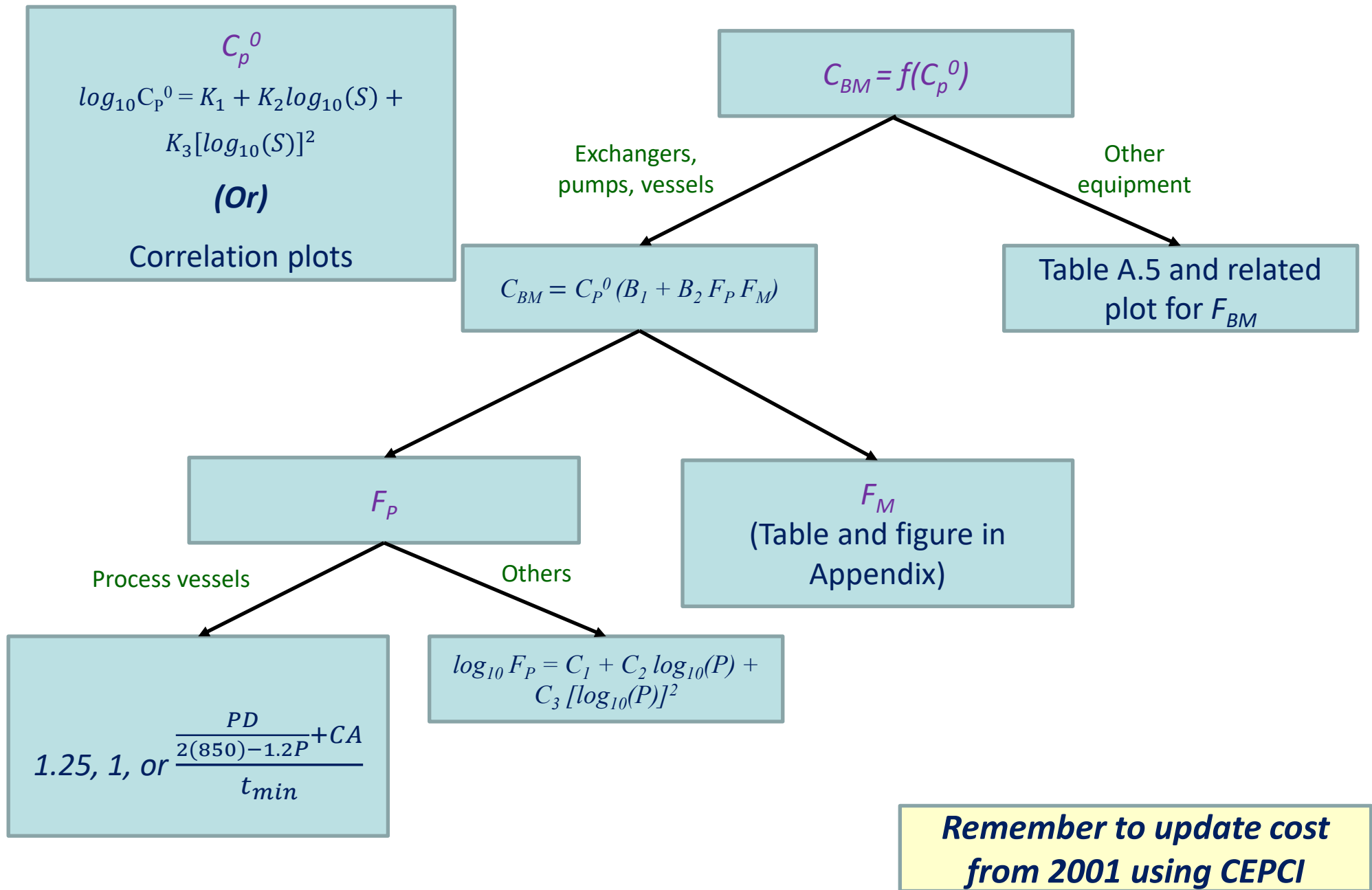
$$P = 100 \text{ barg: } C_{BM} = C_P^0 F_{BM} = C_P^0 (B_1 + B_2 F_P F_M)$$

Capital Cost Estimation – Summary of Procedure

Procedure for Calculating Bare Module Cost

1. Using the correct figure in Appendix A or the data in Table A.1, obtain C_P^0 for the desired piece of equipment => ***purchased equipment cost for base case conditions***
2. Find correct relation for F_{BM}
 - For ***exchangers, pumps and vessels***, use data in Table A.4
 - ✓ Find F_P (equation 3 or 4, data in Table A.2) and F_M (Table A.3, Figure A.18); then calculate F_{BM}
 - For ***other equipment (e.g., compressors)***, use equations in Table A.5
 - ✓ Use Table A.6 and Figure A.19 to find F_{BM}
3. Calculate the bare module cost, $C_{BM} = C_P^0 F_{BM}$
4. Update the cost from 2001 (CEPCI = 397) to the ***present***

Flowchart of the Procedure



Self-study Example

Example 7.13: Six-step Algorithm

Find the bare module cost (in 2011) of a stainless steel tower 3 *m* in diameter and 30 *m* tall. The tower has 40 stainless steel sieve trays and operates at 20 *barg*.

❑ Cost of tower

a) $\text{Volume} = \pi \frac{D^2 L}{4} = 212.1 \text{ m}^3$

=> Purchase cost, $C_p^0(2001) = \$132,500$ using equation (2):

$$\log_{10}(C_p^0) = 3.4974 + 0.4485 \log_{10}(212.1) + 0.1074 [\log_{10}(212.1)]^2$$

$$C_p^0(2011) = \$132,500 \times (582/397) = \$194,240$$

b) From equation (1) and Table A.4, $F_{BM} = 2.25 + 1.82 F_M F_P$

Step 1:

Calculate volume

Step 2:

Calculate base purchase cost with coefficients from Table A.1

Step 3:

Update cost using cost index

Step 4:

Identify correct correlation for bare module factor

c) From equation (4),
$$F_P = \frac{\frac{(20)^3}{(2)(850)-1.2(20)} + 0.00315}{0.0063} = 6.47$$

Step 5:

Calculate pressure factor using appropriate equation

From Table A.3, identification number for stainless steel vertical vessel = 20; from Figure A.18, $F_M = 3.11$

Step 6:

Find material factor

Therefore, $F_{BM} = 38.87$

Step 7:

Calculate bare module cost

d)
$$C_{BM}(2011) = (194,240)(38.87) = \$7,550,300$$

❑ Cost of trays

a) Tray area =
$$\pi \frac{D^2}{4} = 7.0686 \text{ m}^2$$

=> Purchase cost, $C_p^0(2001) = \$4570$ using equation (2):

$$\log_{10}(C_p^0) = 2.9949 + 0.4465 \log_{10}(7.0686) + 0.3961 [\log_{10}(7.0686)]^2$$

$$C_p^0(2011) = \$4570 \times (582/397) = \$6700$$

b) From Table A.5, $C_{BM} = C_p N F_{BM} f_q$, where, $N = 40$ and $f_q = 1.0$ (since number of trays ≥ 20)

From Table A.6, identification number for stainless steel sieve trays = 61; from Figure A.19, $F_{BM} = 1.83$

$$C_{BM}(2011) = (6700)(40)(1.83)(1.0) = \$490,400$$

□ Total cost = $7,550,300 + 490,400 = \$8,040,700$

Total Module and Grass Roots Cost

Total Module and Grass Roots Costs

❑ *Total module cost:*

$$C_{TM} = \sum_i (C_{BM} + C_{Cont} + C_{Fee})_i = 1.18 \sum_i C_{BM,i}$$

❑ *Grass roots plant cost:*

$$C_{GR} = C_{TM} + C_{Site} + C_{Aux} + C_{Off-sites}$$

$$C_{GR} = C_{TM} + 0.5 \sum_i (C_{BM,i})^{for CS and 1 atm}$$

- ❑ Always remember to update the estimated cost to the present/future time using cost index

Self-study Example

Example 7.14: Small Expansion to an Existing Facility

The expansion involves a new distillation column with condenser, reboiler, pumps and other associated equipment. Calculate total module cost for this expansion. *CEPCI has been ignored in the following calculations.*

Equipment	Capacity/Size	Material, Pressure
E-101 Condenser	170 m^2 , Shell & Tube (Floating Head)	Tube: CS, 5 barg Shell: CS, 5 barg
E-102 Reboiler	205 m^2 , Shell & Tube (Floating Head)	Tube: SS, 18 barg Shell: CS, 6 barg
E-103 Product Cooler	10 m^2 , Double Pipe	CS, 5 barg (all)
P-101A & B Reflux Pumps	Shaft Power = 5 kW Centrifugal	CS, Discharge $P = 5$ barg
T-101 Column	2.1 m diameter, 23 m height, 32 sieve trays	CS, 5 barg SS Trays
V-101 Reflux Drum	1.8 m diameter 6 m length, horizontal	CS, 5 barg

❑ Estimate cost of each equipment using data/correlations in Appendix A

■ *For example, E-102 Reboiler*

➤ Purchase cost, $C_p^0 = \$36,900$ using equation (2):

$$\log_{10}(C_p^0) = 4.8306 - 0.8509 \log_{10}(205) + 0.3187 [\log_{10}(205)]^2$$

➤ $F_M = 1.81$ from Figure A.18

➤ $F_p = 1.023$ using equation (3):

$$\log_{10} F_p = -0.00164 - 0.00627 \log_{10}(18) + 0.0123 [\log_{10}(18)]^2$$

➤ Bare module cost, $C_{BM} = C_P^0 [B_1 + B_2 F_P F_M]$ (equation 1)
 $= 36,900 (1.63 + 1.66 \times 1.023 \times 1.81)$
(with B_1 & B_2 from Table A.4)
 $= (36,900)(4.7) = \$173,430$ (here, $F_{BM} = 4.7$)

- Likewise, estimate C_{BM} for each of the other equipment
- Total bare module cost, $\sum_i C_{BM,i} = \$797,000$
- Total module cost, $C_{TM} = 1.18 \sum_i C_{BM,i} = \$940,460 \approx \$940,000$

Concluding Remarks

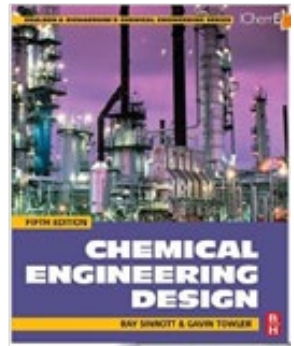
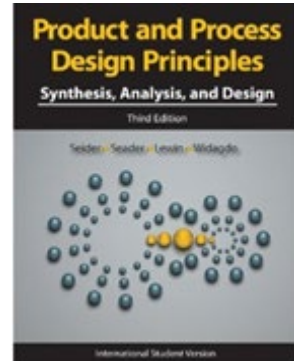
Capital Cost Estimation – Reflections

- ❑ Capital cost estimation
 - Straightforward procedures
 - Requires lot of data/correlations
 - Size, MOC and operating pressure of each equipment

- ❑ Programs for capital cost estimation
 - ***Study Section 7.3.8 on CAPCOST program (5th edition)***
 - Try this program for the Example 7.14 problem

Capital Cost Estimation – Other Programs

- CCEP based capital cost estimation correlations in Seider W.D., Seader J.D., Lewin D.R. and Widagdo S., “Product and Process Design Principles: Synthesis, Analysis, and Analysis”, 3rd edition, John Wiley (2010)
- DFP based on detailed factorial method and cost data in Sinnott R.K. and Towler G., “Chemical Engineering Design” 5th edition, Butterworth and Heinemann (2009)



- DFP and CCEP programs available in **CANVAS-Files**
- Which reference/program should be used?
 - Read the cover story in “Chemical Engineering”, p. 22-29, August 2011, available in **CANVAS-Files**

Cover Story

Evaluating Capital Cost Estimation Programs

Five capital-cost-estimation programs are compared using a set of case studies

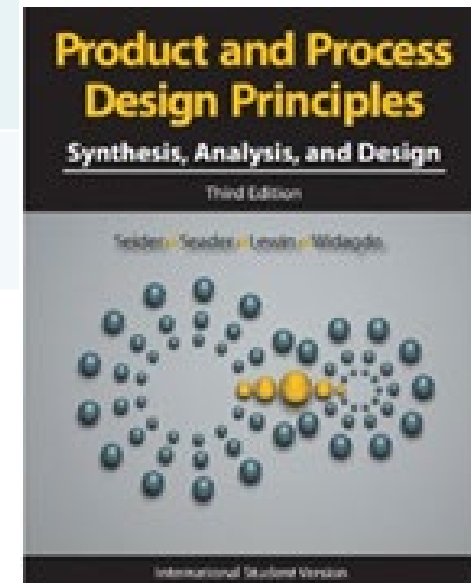
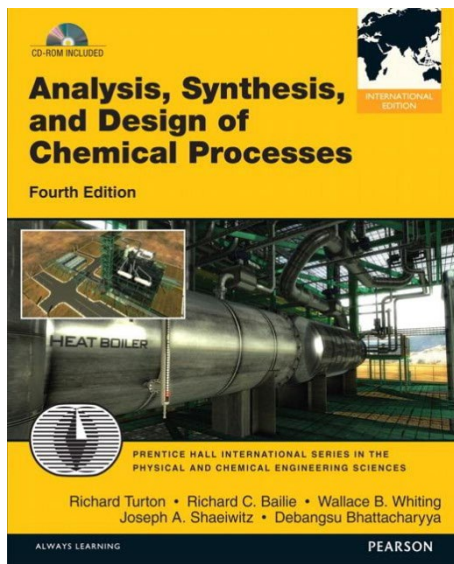
Ying Feng and Gade P. Rangaiah
National University of Singapore

TABLE 1. CLASSIFICATION OF CAPITAL COST ESTIMATES			
AACE International Recommended Practice No. 18R-97 cost estimate classification system			
Class / type	Purpose	Accuracy	Level of project completion
Class 5: Order of magnitude estimates	Initial feasibility study or screening	L: -20 to -50% H: 30 to 100%	0-2%
Class 4: Study or preliminary estimates	Concept study or feasibility	L: -15 to -30% H: 20 to 50%	1-15%
Class 3: Definitive estimates	Budget, authorization or control	L: -10 to -20% H: 10 to 30%	10-40%
Class 2: Detailed estimates	Control or bid/tender	L: -5 to -15% H: 5 to 20%	30-70%
Class 1: Check estimates	Check estimate or bid/tender	L: -3 to -10% H: 3 to 15%	50-100%

Capital Cost Estimation – The Last Word!

- ❑ Fixed capital investment (FCI)
 - Total module cost **OR** grass roots cost
- ❑ Procedures and terminology not standard

Turton et al.	Seider et al.
Bare module cost	Same
Total module cost	???
Grass roots cost	Total depreciable capital
	Total permanent investment



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